

Automated Patient History Summarization

AI Clinical Intelligence — from years of historical reports to an instant, structured summary inside the existing reporting workflow.

1 How It Works

Clinicians routinely face **too much information and too little time**: multiple prior imaging reports, historical findings, follow-up recommendations, and years of accumulated history across different modalities. Reviewing all of it manually is slow, easy to overlook, and inconsistent between readers. The solution replaces that manual mental summarization with an **AI-generated patient summary** that is ready the moment the exam is opened.

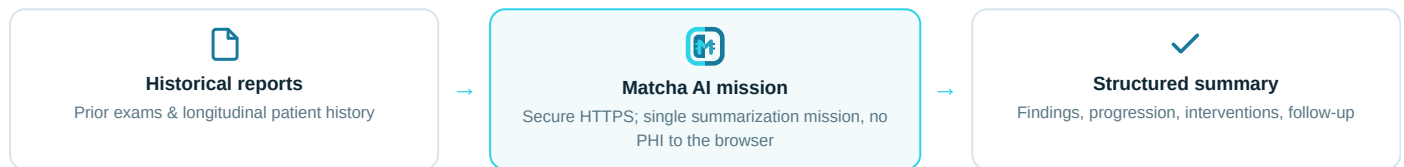
TRADITIONAL WORKFLOW

Search › Open reports › Read each report ›
Build mental summary

AI-ASSISTED WORKFLOW

Open exam › Review AI summary ›
Validate findings › Report

Processing pipeline



What the summary contains

- Significant findings and active problems / risk factors
- Disease progression across studies over time
- Prior interventions and procedures
- Follow-up recommendations and historical trends

AI processing strategy

- **Prefetch generation** — triggered when an order is received.
- **Final refresh** — triggered when the report is verified by a physician.
- **Caching** — reuses results to cut tokens, cost, and latency.
- Summaries are ready before the clinician needs them — no on-demand wait.

Standards-based clinical intelligence

USCDI v4 — U.S. Core Data for Interoperability

Standardized data elements: problems & conditions, procedures, medications, allergies, clinical notes, diagnostic reports.

C-CDA — Consolidated Clinical Document Architecture

Industry-standard document structure: patient history, encounter information, findings, and recommendations.

Embedded, not bolted on

One-click access inside the existing reporting experience — **no workflow disruption**, a clinically focused presentation, and a single summarization mission that supports all exam types, specialties, and customer environments.

Where the Value Comes From

Faster review, better clinical context, and lower cost — delivered without changing how clinicians already work.

Seconds

Summary ready on exam open
(prefetched)

Years → 1

History condensed to one structured view

1

Mission for all exam types & specialties

0

Extra clicks / workflow disruption

2 The Value

🕒 Faster review

Clinicians skip searching and re-reading prior reports. The summary is prefetched and waiting, so review starts immediately instead of after several seconds of on-demand generation.

🎯 Better clinical context

Significant findings, disease progression, and prior interventions are surfaced together, reducing the risk that important historical findings are overlooked.

👤 Improved consistency

A standards-based structure (USCDI v4 / C-CDA) gives every reader the same organized view, reducing inconsistent understanding between clinicians.

⬆️ Enhanced productivity

Less time rebuilding history mentally means lower reporting workload and more capacity for the exams that need it — without adding steps to the workflow.

💰 Cost efficiency & scale

- Caching reduces repeated AI requests and token consumption.
- Best-fit Matcha model chosen on quality, speed, and cost.
- Scales across large, multi-site customer environments.
- Platform flexibility: Matcha today, open-source models optional.

🛡️ Safe & easy to adopt

- Server-side API key; the browser never talks to Matcha directly.
- PHI screening before requests leave the web tier.
- Single mission → simpler deployment & maintenance.
- Runs on current v5.21 customers; future rollout on v5.22.

Business outcome

Clinicians move **from historical reports to actionable clinical intelligence**: better context, smarter reporting, and AI-powered workflows — with faster customer feedback, a larger pilot audience, and reduced upgrade barriers.

Faster review

Better context

Improved consistency

Lower cost per summary

No workflow disruption

Standards-based